

Endo-HJEPa: A Hierarchical Joint-Embedding World Model for Unified Endoscopic Video

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Abstract

Purpose. Minimally invasive diagnosis and therapy are increasingly performed through an endoscope, yet onboard intelligence remains almost entirely *reactive*: systems recognise the present frame but cannot anticipate how the scene will evolve, or how it would change under a candidate camera action. Safe guidance, collision warning, and any future scope assistance require a *world model* of endoscopic dynamics. Endoscopic appearance is dominated by unpredictable factors (specular highlights, fluid, smoke, peristalsis), so pixel-generative world models spend capacity on what cannot be predicted. We argue that the right object to predict is a *plannable representation*, not the next frame. **Methods.** We present Endo-HJEPa, a hierarchical joint-embedding predictive architecture trained as one domain-conditioned world model across laparoscopy, flexible GI endoscopy, and bronchoscopy. A shared frozen V-JEPa 2 ViT-L encoder maps clips to dense spatio-temporal tokens. Three predictors operate at increasing temporal abstraction: a causal autoregressive L1 forecaster (short-horizon residual dynamics), a coarse L2 predictor (mid-horizon anatomy/phase), and an action-conditioned L3 predictor over a vector-quantised residual codebook. A contrastive energy head scores transition plausibility and drives sampling-based latent model-predictive control. Residual anchoring, uncertainty-weighted multi-task loss, VICReg anti-collapse, and endoscopic token weighting (specular down, instrument up) complete the training objective. No pixels are decoded. **Results.** On a video-level protocol over 1,707 sequences (1,067,026 frames, 19 datasets), the causal L1 forecaster reaches 0.978 cosine at horizon 4 versus 0.916 (persistence), 0.974 (GRU) and 0.971 (Mamba); Wilcoxon $p < 10^{-80}$ on 750 held-out clips. The same protocol beats persistence on every val dataset with $n \geq 5$ (largest margins on Stereo_Lap +0.093 and ION bronchoscopy +0.071). Full H-JEPa is the only compared model that plans: latent MPC is closer to the goal than persistence on 98.0% of clips, and the energy head flags near-wall SCARED transitions (AUC 0.682, Spearman -0.471). On CholecT50 (official split, 3 seeds) frozen V-JEPa 2 is the strongest *video* backbone for phase recognition (0.704 ± 0.006); domain adaptation lifts instrument mAP from 0.406 to 0.485. Two negative results are reported as findings: emergent latent actions are not grounded without encoder-level supervision, and goal-directed navigation to arbitrary anatomical targets remains hard (10% reach). **Conclusions.** A hierarchical, energy-based JEPa can be

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trained and evaluated as a unified cross-orifice endoscopic world model, providing forecast, planning, and a physically correlated uncertainty signal without pixel generation. Latent actions are a planning device, not interpretable surgical actions. Code and protocols are released.

Keywords: world model, joint-embedding predictive architecture, endoscopy, representation learning, latent planning, self-supervised learning

1. Introduction

1.1. Clinical context: endoscopy needs anticipation, not only recognition

Minimally invasive diagnosis and therapy are increasingly delivered through an endoscope—laparoscopes in the abdomen, flexible scopes in the gastrointestinal tract, and bronchoscopes in the airway. Across all three orifices, safe operation depends on a clinician’s ability to *anticipate*: how tissue will deform under contact, how a lumen will open as the scope advances, where an instrument will move, and when the camera is about to collide with a wall or lose the view. Current endoscopic AI is almost entirely *reactive*. It recognises the present frame—surgical phase, instrument presence, anatomy—but cannot forecast what comes next, nor what a candidate camera adjustment would produce. A model that predicts how an endoscopic scene will evolve, and how it would evolve under a candidate action, is a prerequisite for anticipatory guidance, collision warning, and any future assisted scope control. That is the clinical target of this paper: not a stronger phase classifier, but a *world model* that can be queried about the future.

1.2. The technical problem: structure is predictable, appearance is not

A world model for endoscopy faces a specific asymmetry. The *structure* of endoscopic video—lumen geometry, camera advance, instrument trajectories—follows smooth physical dynamics and is strongly predictable. The *appearance*—specular highlights, fluid, smoke, mucosal micro-texture, peristalsis—is high-entropy and largely unpredictable. The dominant world-model paradigm, pixel generation (diffusion world-action models, Genie-style token video), must spend capacity forecasting this unpredictable appearance. That both wastes capacity and ties the model to a pixel metric that is dominated by glare and fluid rather than by the quantities that matter for planning. Joint-embedding predictive architectures (JEPA) offer the alternative we adopt: predict in a learned representation space, where unpredictable appearance residual can be ignored by design, and where the predicted quantities are exactly the plannable ones.

Key insight: predict plannable representations, not the next frame.. If a frame is written as structure plus appearance, any pixel predictor is lower-bounded by the appearance dispersion, regardless of how well structure is modelled (Section 3.8). A representation that is sufficient for structure and invariant to appearance has no such bound. Endoscopy is an unusually clean instance of this split: the camera moves through a deformable lumen under clinician control (predictable), while highlights and fluid dominate the pixel residual (unpredictable). The design consequence is that a world model should (i) forecast in encoder space, (ii) separate fast dynamics from slow anatomy, (iii) score which futures are plausible, and (iv) plan by rolling those futures—without ever decoding pixels.

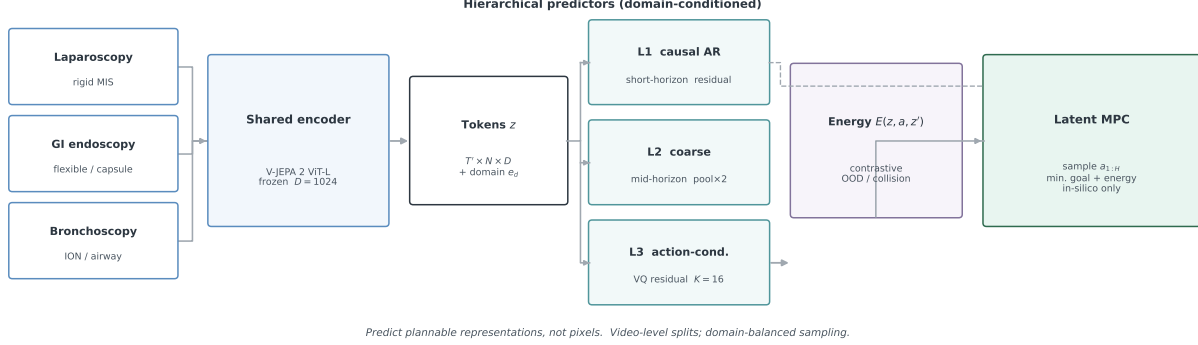


Figure 1: Overview of Endo-HJEPA. Three orifice streams share a frozen V-JEPA 2 ViT-L encoder. Domain-conditioned L1/L2/L3 predictors forecast future tokens at increasing temporal abstraction. A contrastive energy head scores transition plausibility and drives latent MPC. Boxes hold only the module name and the governing variable; no pixels are decoded. When a goal latent is unavailable, the system reduces to L1 forecast plus energy-based rejection.

1.3. Gap

Despite rapid progress in surgical video understanding, two gaps remain. **(i) Single scale.** Existing surgical JEPA and video-pretraining systems (SurgRec-JEPA, JHU-VPT, VideoMAE) learn a single temporal predictor and do not separate fast tissue/tool/camera dynamics from slow anatomical and procedural structure. LeCun’s hierarchical JEPA (H-JEPA) argues such separation is essential for long-horizon prediction and planning, but it has not been instantiated for endoscopy. **(ii) Per-orifice fragmentation.** Endoscopic models are built per orifice—laparoscopy, GI endoscopy, and bronchoscopy are treated as separate foundation-model problems—even though all three share the same underlying physics of a camera advancing through a deformable lumen under clinician control. A single domain-conditioned world model is both a stronger scientific claim and a more data-efficient one, but it requires evidence that such sharing is beneficial, or necessary.

1.4. Our approach

We instantiate this idea in **Endo-HJEPA** (Figure 1). A shared frozen V-JEPA 2 ViT-L encoder maps clips from all three orifices to dense tokens. Three domain-conditioned predictors operate at increasing temporal abstraction: L1 (causal autoregressive, residual, short horizon), L2 (temporally pooled, mid horizon), and L3 (action-conditioned over a residual codebook). A contrastive energy head scores (z, a, z') transitions and is used both as an out-of-distribution prior and as the cost in sampling-based latent MPC. The encoder is frozen by default so that every dynamics ablation shares identical latents; an optional last-block adaptation is evaluated only on recognition probes. No decoder is trained. The evidence is scoped as follows: forecast and scale are measured on a 6,000-clip causal-L1 protocol; planning, energy, and physical probes use the full H-JEPA trained at 2,000 clips; recognition uses frozen versus adapted V-JEPA 2 on the official CholecT50 split. These protocols are never mixed in a single table.

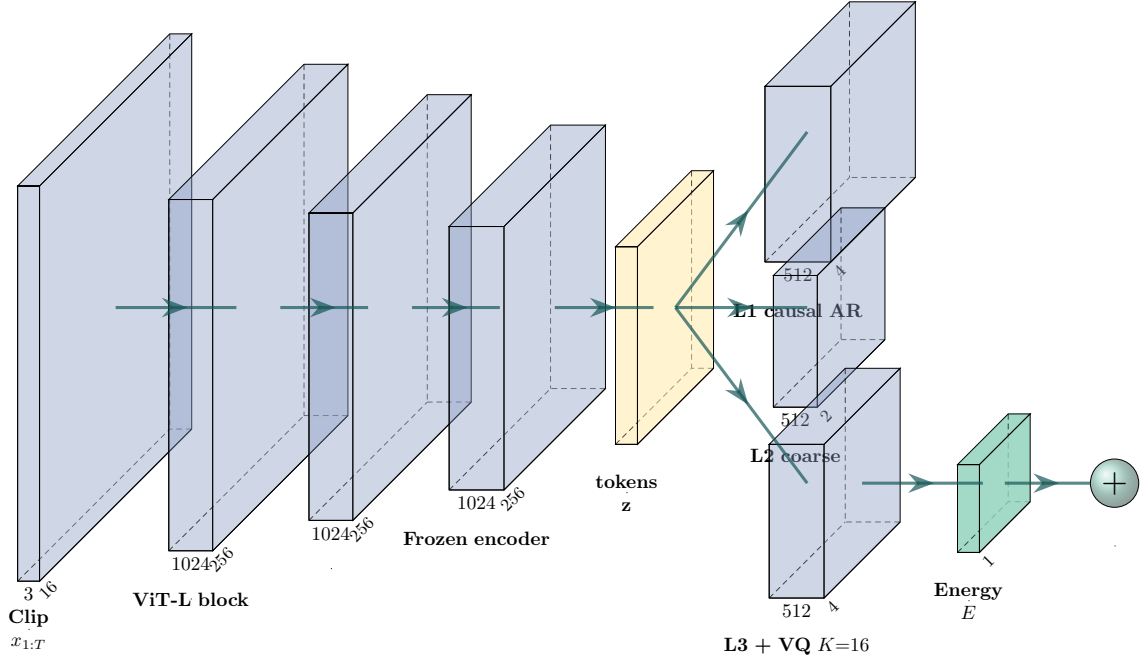


Figure 2: PlotNeuralNet rendering of the same stack: input clip, frozen ViT-L blocks, token tensor, L1/L2/L3 predictors, energy head, and latent-MPC aggregator. Colours are muted (print-safe blue/teal); block sizes encode relative tensor rank, not a CNN.

1.5. Contributions and significance

1. **A hierarchical, energy-based JEPA world model for unified cross-orifice endoscopy.** We instantiate H-JEPA (short-horizon residual dynamics, mid-horizon anatomy, action-conditioned planning with an energy prior) on the official V-JEPA 2 ViT-L encoder, trained and evaluated across laparoscopy, GI endoscopy, and bronchoscopy with one domain-conditioned dynamics model.
2. **A causal-autoregressive latent forecaster.** The *form* of the latent forecaster matters more than its size: a GPT-style causal Transformer surpasses a strong GRU and a Mamba/SSM baseline, whereas a parallel query-token Transformer of the same capacity does not.
3. **A physically correlated uncertainty signal.** The contrastive energy head flags out-of-distribution transitions and correlates with camera-to-tissue distance on SCARED (near-wall AUC 0.682), not only with latent self-consistency.
4. **A leakage-free, per-dataset protocol.** Video-level splits, domain-balanced sampling, per-dataset forecast decomposition, reproducible external baselines, and paired Wilcoxon tests with Holm correction, released as executable code.

We also report two characterised negative results: self-supervised latent actions are not semantically or physically grounded without encoder-level supervision, and zero-shot cross-orifice transfer fails but is recoverable with a 32-shot domain token. Together these provide evidence that unified multi-orifice training is necessary, not merely convenient. We do *not*

claim the first surgical world model, pixel-level video generation, interpretable latent actions, uniform recognition SOTA, or any clinical efficacy endpoint.

Significance. Endo-HJEPA reframes endoscopic AI from reactive recognition to anticipatory world modelling. Three properties matter clinically: anticipation (forecast is the primitive behind collision warning), a grounded uncertainty signal (energy tracks physical proximity), and unification (one domain-conditioned model serves three orifices where labelled video is scarce). All planning results are in-silico.

2. Related work

We position Endo-HJEPA along five axes: the world-model lineage it builds on, the surgical/endoscopic representation learning it extends, the surgical world models it differs from, the general video/image backbones we benchmark against, and the endoscopic datasets that anchor our evaluation.

2.1. World models and joint-embedding prediction

The idea of a learned world model for prediction and planning originates with Ha and Schmidhuber’s World Models [1] and the PlaNet latent-dynamics planner [2], and was scaled to diverse control domains by Dreamer [3]. These models predict a learned latent state and plan by rolling it forward, but they are trained with a reconstruction or reward signal and target control, not video understanding. Energy-based formulations [4] provide the formal basis for scoring state compatibility without a generative decoder. The joint-embedding predictive architecture (JEPA) of LeCun’s position paper [5] argues that predicting in representation space—rather than pixels—is the key to capturing predictable structure while ignoring unpredictable appearance; I-JEPA [6] realised this for images, V-JEPA [7] for video, and V-JEPA 2 [8] added scale and the action-conditioned (V-JEPA 2-AC) planning variant. Generative interactive environments such as Genie [9] take the opposite, pixel-generative route. A common limitation of these works, for our purpose, is that they are either single-scale (no temporal hierarchy), pixel-generative (capacity spent on unpredictable appearance), or evaluated on generic/control video rather than endoscopic dynamics. Endo-HJEPA adopts the JEPA principle and adds the hierarchy and energy prior that the surgical setting lacks.

2.2. Self-supervised surgical and endoscopic video representation

Self-supervised pretraining has become the default for surgical video. SurgRec-JEPA [10] and JHU-VPT [11] apply JEPA-style and video pretraining to surgical data and report strong downstream transfer, and endoscopic foundation models such as Endo-FM [12] scale self-supervised pretraining to large endoscopic corpora. These methods produce a single-scale representation for *recognition* (phase, tools, segmentation); none of them build a dynamics model that can forecast or plan. Endo-HJEPA shares their self-supervised, frozen-encoder efficiency but adds the temporal hierarchy and the action-conditioned planning layer that recognition-oriented SSL omits.

2.3. Surgical and endoscopic world models

World models tailored to surgery are emerging. The Surgical Vision World Model [13] generates pixel-controllable surgical video; SurgWorld [14] learns latent actions for cataract surgery; EndoWAM [15] applies diffusion world-action modelling to endoscopic navigation; and Surgical WAM [16] couples generative video with robot action chunks for closed-loop control. These are predominantly *pixel-generative* or *single-task*; they neither separate temporal scales nor provide an explicit uncertainty/energy signal for safe planning. Our contribution is orthogonal and complementary: a hierarchical JEPA with energy-driven latent MPC that predicts representations, not pixels.

2.4. General video and image representation backbones

For the representation evaluation we benchmark against the strongest *reproducible* general-purpose encoders. For video: VideoMAE [17] (Kinetics masked autoencoding), TimeSformer [18] (divided space-time attention), and ViViT [19] (factorised space-time); state-space video models such as VideoMamba [20] and the underlying Mamba block [21] are the recurrent/SSM reference for dynamics. For images: the ImageNet-supervised ViT [22] and DINOv2 self-supervised features [23], pooled over time. None of these provide a world-model predictor or a planning/energy capability, which is the gap Endo-HJEPA fills; §4 measures them on the same CholecT50 protocol.

2.5. Endoscopic datasets and physical supervision signals

Our evaluation spans the standard endoscopic corpora: Cholec80 (phase + tool presence) [24], CholecT50 (action triplets + phase) [25, 26], EndoVis 2017/2018 (instrument segmentation) [27], CholecSeg8k (semantic segmentation) [28], endoscapes (segmentation + critical view of safety) [29], the GI corpora HyperKvasir [30] and Kvasir-Capsule [31], and the pose/depth corpora SCARED [32] and C3VD [33], with STIR [34] for deformable point tracks. To our knowledge, using SCARED/C3VD camera poses and STIR point tracks as *physical anchors* for latent actions, energy, and deformation—rather than as reconstruction targets—is a novel use of these supervision signals within a JEPA world model.

Summary.. Across the five axes, the field offers strong single-scale SSL representations, pixel-generative surgical world models, and general video/image backbones—but no hierarchical, energy-based JEPA world model trained and evaluated as a unified cross-orifice endoscopic system with a physically grounded uncertainty signal. That is the gap Endo-HJEPA occupies.

3. Method

Core idea.. Endo-HJEPA is a world model that answers two queries without generating pixels: (i) given a history of endoscopic tokens, what is the likely future representation; (ii) given a candidate latent action, is the implied transition plausible, and does it approach a goal representation. The first query is L1/L2 forecast; the second is L3 plus energy plus latent MPC. Everything else in this section—residual anchoring, uncertainty weighting, VICReg, specular/instrument token weights, STIR chamfer—exists to make those two queries stable on endoscopic video. Figure 1 is the data flow; Figure 2 is the tensor-level stack; Table 1 lists symbols.

Table 1: Principal notation (Section 3).

Symbol	Meaning
$x_{1:T}, d$	Input clip of T frames; orifice $d \in \{\text{laparo, gi, bronch}\}$
$f_\theta, z_{1:T'}$	Frozen encoder; dense tokens $z \in \mathbb{R}^{T' \times N \times D}$
T', N, D	Tubelet steps ($T/2$), spatial tokens (256), width (1024)
e_d	Learnable domain embedding
H, H_{hist}	Forecast horizon and history (default 4)
$a_t \in \{1, \dots, K\}$	Discrete latent action ($K=16$)
$r_t = z_{t+1} - z_t$	Residual used by VQ and residual prediction
$E(z, a, z')$	Scalar transition energy (low = plausible)
s_i	Learnable log-variance of loss term i
$z^\star$	Goal latent for MPC
w	Specular / instrument tubelet weights

3.1. Problem formulation and assumptions

A clip $x_{1:T}$ from orifice d is mapped by a frozen encoder f_θ to

$$z_{1:T'} = f_\theta(x_{1:T}) \in \mathbb{R}^{T' \times N \times D}, \quad T' = T/2, \quad N = 256, \quad D = 1024. \quad (1)$$

The world model is a collection of predictors g_ϕ on these tokens. It never decodes pixels. We factor g_ϕ into three temporal scales, mirroring H-JEPA: L1 forecasts $\hat{z}_{t+1:t+H}$ (tissue, tool, camera); L2 forecasts a stride-2 pooled stream (anatomy / phase); L3 forecasts the consequence of a discrete action a_t . A scalar energy $E(z_t, a_t, z_{t+1})$ scores transition plausibility. Planning is sampling-based MPC:

$$\hat{a}_{1:H} = \arg \min_{a_{1:H}} \left\| \hat{z}_{t+H} - z^\star \right\|_2^2 + \sum_{k=0}^{H-1} E(\hat{z}_{t+k}, a_{t+k}, \hat{z}_{t+k+1}). \quad (2)$$

Assumptions. (i) Endoscopic dynamics are approximately Markovian in the learned latent space at the tubelet timescale. (ii) The frozen V-JEPA 2 encoder is more predictable for structure than for appearance (tested by the pixel-generation contrast in §4.2). (iii) Orifice domains share enough low-level dynamics that a domain-conditioned predictor can serve all three, while differing enough that naive zero-shot transfer fails (tested in §4.4).

3.2. Shared encoder and endoscopic token weighting

We use facebook/vjepa2-vitl-fpc64-256 (ViT-L, patch 16, tubelet 2, input 256^2), frozen by default. An optional flag unfreezes the last K blocks for recognition-oriented domain adaptation; that path is never mixed into the forecast tables. High-luminance, low-saturation tubelets (glare, fluid highlights) are weakly predictable and are down-weighted to a floor $w_{\min}=0.25$. When EndoVis instrument masks are available, instrument tubelets are up-weighted by $1 + \beta m_{\text{inst}}$. The resulting weights w multiply the L1 residual; they do not change the encoder.

3.3. Hierarchical predictors

All predictors are Transformer-based. A shared domain embedding e_d is added to the history so that one set of weights serves three orifices.

182 *L1, causal autoregressive (default pooled forecaster)..* A GPT-style causal Transformer pre-
 183 dicts the next pooled token from the causal context and feeds the prediction back for H
 184 steps. Let $h_t = \text{in}(z_t) + e_d + p_t$ be the projected history with positional encoding p_t , and let
 185 M be the upper-triangular causal mask. One rollout step is

$$\hat{z}_{t+1} = z_t + \text{head}(\text{Transformer}(h_{\leq t}; M))_t, \quad (3)$$

186 i.e. the network predicts a *residual* from the last observed (or last predicted) token. This
 187 matches the near-Markov, smooth structure of endoscopic camera motion. Section 4.2 shows
 188 that this inductive bias, not parameter count, is what lets the Transformer surpass a GRU:
 189 a parallel query-token Transformer of the same size does not.

190 *L1, dense spatio-temporal..* For spatially resolved prediction a factorised block applies tem-
 191 poral attention per spatial site, then spatial attention per tubelet step, over all $N=256$ tokens.
 192 Learned positional embeddings are added along both axes; H query steps are appended along
 193 time. This captures global camera motion that independent per-site prediction cannot rep-
 194 resent. Dense L1 is trained jointly with the pooled head so that the pooled evaluation path
 195 is not an untrained leftover.

196 *Residual (delta) prediction..* Consecutive endoscopic latents are highly correlated. Every
 197 predictor therefore forecasts the change from the last observed token,

$$\hat{z}_{t+k} = z_{T'} + f_\phi(\cdot)_k, \quad (4)$$

198 rather than an absolute future. The model can only improve on persistence; without residual
 199 anchoring a 70M predictor is high-variance and can fall below persistence (Section 4.8).

200 *L2, coarse mid-horizon..* L1 tokens are average-pooled along time with stride 2; a second
 201 Transformer predicts the coarser future. L2 adds little to short-horizon pooled cosine; its
 202 role is a seconds-scale handle for anatomy and phase.

203 *L3, action-conditioned..* Residuals $r_t = z_{t+1} - z_t$ are vector-quantised into a codebook of
 204 $K=16$ entries (straight-through estimator, commitment loss). The L3 predictor conditions
 205 on the resulting discrete ids. These ids are a *planning device*: they index visual change, not
 206 surgical verbs (Section 4.5).

207 *Energy head..* $E(z, a, z')$ is an MLP on $[z; \text{emb}(a); z']$, trained contrastively with a rolled
 208 negative z^- :

$$\mathcal{L}_E = \text{softplus}(E(z_t, a_t, z_{t+1}) - E(z_t, a_t, z^-)). \quad (5)$$

209 Low energy marks in-distribution, predictable futures. High energy is used as a reject /
 210 collision-risk prior.

211 *Latent MPC..* We sample $N_{\text{mpc}}=32$ action sequences of length H , roll L3, and select the
 212 sequence minimising (2). This is performed entirely in latent space and is in-silico only.

3.4. Objective

Each loss term $\mathcal{L}_i$ carries a learnable log-variance s_i (uncertainty weighting), contributing $e^{-s_i}\mathcal{L}_i + s_i$. This stops dense L1 from being diluted by L2/L3/energy. Dense JEPA prediction can collapse; we add a VICReg-style regulariser on $\hat{z}$,

$$\mathcal{L}_{\text{vr}} = \frac{1}{D} \sum_j \max(0, 1 - \text{std}(\hat{z}_{:,j})) + \frac{1}{D} \sum_{i \neq j} \text{Cov}(\hat{z})_{ij}^2. \quad (6)$$

The full objective is

$$\mathcal{L} = \sum_{i \in \mathcal{I}} (e^{-s_i} \mathcal{L}_i + s_i) + \lambda_{\text{vr}} \mathcal{L}_{\text{vr}} + \lambda_s \mathcal{L}_{\text{smooth}} + \lambda_{\text{STIR}} \mathcal{L}_{\text{STIR}}, \quad (7)$$

with $\mathcal{I} = \{\text{L1}, \text{L1}_{\text{pool}}, \text{L2}, \text{L3}, E, \text{commit}\}$. Persistence (copy last latent) is a mandatory baseline for every forecast table.

3.5. STIR point-track regulariser

STIR provides IR start/end segmentations rather than dense tracks. We sample up to 64 points from each mask, map them onto the token grid, and apply a symmetric chamfer between tokens at those sites at the first and last timestep:

$$\mathcal{L}_{\text{STIR}} = \text{Chamfer}(z_0[p_{\text{start}}], z_{T'-1}[p_{\text{end}}]). \quad (8)$$

This constrains L1 deformation in representation space, with no pixel regression.

3.6. Physical alignment of latent actions

SCARED stores a 4×4 camera pose per frame; C3VD provides `pose.txt`. The relative pose $T_t^{-1}T_{t+1}$ is mapped to a 6D twist. We quantify grounding by (i) normalised mutual information between k -means clusters of latent residuals and of twists, against a random-id chance baseline, and (ii) a closed-form linear probe from residual to twist (MAE, R^2). This is a *probe*, not a training loss, unless encoder-level verb supervision is explicitly enabled.

3.7. Training procedure

Clips are enumerated from a sequence-level manifest with video-level splits (hash of `dataset::sequence_id`); no clip crosses a split boundary. Sampling is domain-balanced by round-robin so laparoscopy does not dominate GI/bronchoscopy. Dense tokens are cached once per encoder; L1, full H-JEPA, GRU, and Mamba train on the *same* cached latents. Optimisation: AdamW (lr 3×10^{-4} , weight decay 0.01), cosine schedule with 5% warmup, gradient clip 1.0. Full H-JEPA: 69.7M trainable parameters; GRU: 2.63M; Mamba: 1.58M.

3.8. Why representations, not pixels

Write a frame as $x = (s, n)$: structure s (lumen, pose, instruments) with $s_{t+1} = g(s_t) + \varepsilon$ and $\mathbb{E}\|\varepsilon\|_2^2 = \sigma_s^2$ small; appearance n (highlights, fluid) drawn independently of the past with dispersion σ_n^2 . A decoder $r(s, n)$ is Lipschitz in n with constant L_n .

Algorithm 1 Endo-HJEPa training and latent MPC (no pixel loss).

- 1: **Training:** for each domain-balanced clip $x_{1:T}$ with domain d :
 - 2: $z_{1:T'} \leftarrow f_\theta(x_{1:T}); \quad w \leftarrow \text{specular/instrument weights}$
 - 3: $\mathcal{L}_1 \leftarrow \text{SmoothL1}(\text{causal-L1}(z_{\text{hist}}, e_d), z_{\text{fut}})$ (residual)
 - 4: $\mathcal{L}_2 \leftarrow \text{SmoothL1}(\text{L2}(\text{pool}(z_{\text{hist}}), e_d), \text{pool}(z_{\text{fut}})[::2])$
 - 5: $\text{ids, commit} \leftarrow \text{VQ}(z_{t+1} - z_t); \quad \mathcal{L}_3 \leftarrow \text{SmoothL1}(\text{L3}(z_{\text{hist}}, \text{ids}, e_d), z_{\text{fut}})$
 - 6: $\mathcal{L}_E \leftarrow \text{contrastive energy (5)}$
 - 7: $\mathcal{L} \leftarrow \sum_i e^{-s_i} \mathcal{L}_i + s_i + \lambda_{\text{vr}} \mathcal{L}_{\text{vr}}; \text{AdamW on } \phi$
 - 8: **Planning (in-silico):** given z_{hist} , goal z^* , domain d :
 - 9: sample N_{mpc} action sequences; roll L3; return arg min of (2)
-

Proposition 1 (pixel prediction is appearance-limited). Any next-frame predictor $\hat{x}_{t+1} = \hat{r}(x_{1:t})$ satisfies $\mathbb{E}\|x_{t+1} - \hat{x}_{t+1}\|_2^2 \geq \frac{1}{2} L_n^2 \sigma_n^2$: the Bayes predictor of independent appearance is its mean, so the pixel error is lower-bounded by appearance dispersion regardless of how well g is modelled.

Proposition 2 (representation prediction is structure-limited). If E is sufficient for structure and invariant to appearance, $E(s, n) = E(s)$, then a JEPA predictor on $z = E(x)$ has error $O(\sigma_s^2)$ with *no* dependence on σ_n^2 .

These statements formalise the design choice; they do not by themselves prove downstream superiority. That is tested in §4.2, where a next-frame CNN (PSNR 17.9 dB) and a conditional DDPM (4.9 dB) both lose to copy-last (29.6–30.4 dB).

4. Experiments

Goal. The experiments test the two queries of Section 3: can Endo-HJEPa forecast plannable endoscopic representations better than persistence and strong recurrent/SSM baselines, and can L3+energy+MPC plan and reject in a way that is physically meaningful. They are organised by six research questions that map onto the contributions, not by dataset dump. Unless marked “lit.”, every number is measured from a named JSON under `outputs/` and copied into `docs/endohjepa/verified_metrics.json`; literature rows are never mixed into a measured comparison. Forecast tables use the 6,000-clip causal-L1 protocol; planning / energy / pose tables use the 2,000-clip full H-JEPA; recognition uses the official CholecT50 split. These protocols are labelled in every caption.

Organization. Section 4.1 states data, metrics, baselines, and implementation. Section 4.2–4.2 report forecast and scale (C1). Section 4.3 reports planning and energy (C2). Section 4.4 reports cross-orifice transfer (C3). Section 4.5 reports action grounding (negative). Section 4.6 reports recognition probes (encoder quality, not the headline claim). Section 4.7 is the per-dataset decomposition. Section 4.8 isolates design choices.

4.1. Datasets, metrics, baselines, implementation

Datasets. Table 2 lists all 19 local corpora (1,707 sequences, 1,067,026 frames). Splits are video-level (hash of `dataset::sequence_id`): train 1,328 / val 188 / test 191 sequences. TrackVes and SurgT contribute sequences but not decoded frames (negative frame counts

Table 2: Dataset census (video-level split). Forecast uses decoded RGB clips; pose/depth/mask columns mark physical probes, not the forecast loss.

Domain	Dataset	Seq.	Frames	Role
GI	HyperKvasir [30]	768	96,194	forecast
GI	Kvasir-Capsule [31]	234	317,609	forecast
GI	Kvasir-Instrument	1	590	forecast
GI	C3VD [33]	1	1,379	pose probe
Laparo	Stereo_Lap	44	185,344	forecast
Laparo	endoscapes [29]	11	153,899	forecast
Laparo	CholecT50 [25]	50	100,863	forecast + phase/tools
Laparo	Cholec80-Boxes [24]	5	15,691	forecast (v41–45)
Laparo	CholecSeg8k [28]	1	8,080	semantic (not main)
Laparo	EndoVis 2017/2018 [27]	13	5,701	forecast + masks
Laparo	SCARED [32]	76	1,855	pose + depth
Laparo	STIR [34]	360	3,520	deformation
Laparo	tracking / own / EndoNeRF	67	12,887	forecast / tracking
Bronch	ION_bronch	76	163,414	forecast (private)
Total		1,707	1,067,026	

in the census); they are excluded from latent forecast. Full Cholec80 is not local (CAMMA request; only videos 41–45 plus CholecT50). C3VD is restricted to `cecum_t1_a` by Drive quota. ION bronchoscopy is private and stored only as anonymised `case_XXX`.

Metrics.. Latent forecast is cosine similarity and MSE between predicted and target *pooled* tokens at horizon h ,

$$\cos_h = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \frac{\langle \hat{z}_{t+h}^{(v)}, z_{t+h}^{(v)} \rangle}{\|\hat{z}_{t+h}^{(v)}\|_2 \|z_{t+h}^{(v)}\|_2}, \quad (9)$$

averaged over video-level val clips $\mathcal{V}$. Higher cosine / lower MSE is better. Planning reach is the fraction of clips on which latent MPC is closer to the goal latent than persistence. Energy quality is the fraction of clips on which planned actions have lower energy than random actions, plus near-wall AUC on SCARED depth. Recognition uses video-level linear-probe phase accuracy and instrument mAP (mean $\pm$ std over 3 seeds). Action grounding uses NMI versus a random-id baseline.

Baselines.. Dynamics: persistence, GRU, Mamba/SSM, query-token L1. Representation: ImageNet ViT, DINOv2, VideoMAE, TimeSformer, ViViT, all on the same official CholecT50 split. Pixel contrast: next-frame CNN and conditional DDPM. V-JEPA 2-AC and SurgRec-JEPA have no public endoscopic-planning checkpoints; we cite their papers and do not invent a numeric comparison.

Statistics.. Paired bootstrap (1,000 resamples) and Wilcoxon signed-rank tests against GRU and Mamba, Holm-corrected across horizons, on 750 video-level clips.

Implementation.. Single GPU; encoder cached once; AdamW as in Section 3. Hyperparameters: hidden 512, 8 heads, 4 layers, dropout 0.1, VICReg weight 0.1, $K=16$, history/horizon 4/4. Varying depth (2–6), width (256–768) and lr (10^{-4} – 10^{-3}) changes 6k-clip cosine by <0.002 .

Table 3: Video-level latent forecast at $h=4$ (6,000-clip training, shared cache). Source: `outputs/scale_6000_*/val_metrics.json`.

Model	cos $h=4$	MSE $h=4$
Persistence	0.916	0.532
Endo-HJEPA (query-token L1)	0.936	0.408
Mamba / SSM dynamics	0.971	0.180
GRU dynamics	0.974	0.161
Endo-HJEPA (causal L1)	0.978	0.139

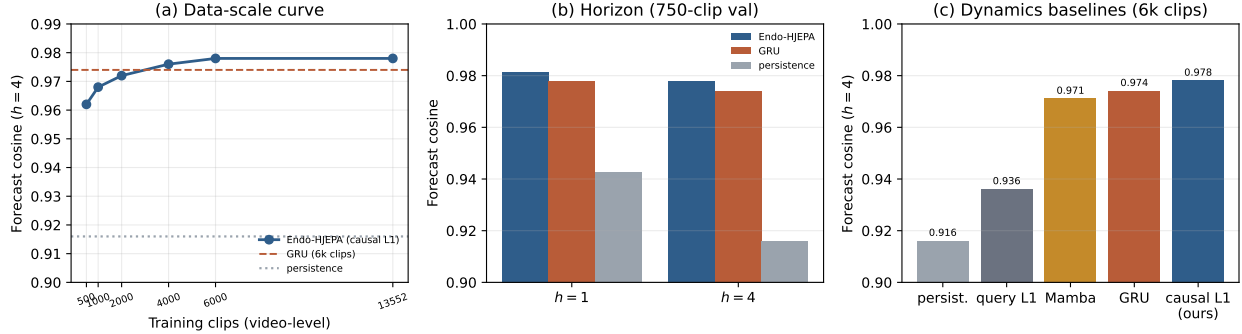


Figure 3: (a) Forecast cosine versus training clips; GRU and persistence are 6k-clip references. (b) Horizon $h=1, 4$ on the 750-clip val set used for Wilcoxon tests (Table 3 uses $h=4$ pooled cosine 0.978). (c) Same 6k-clip protocol as Table 3. Numbers are from `verified_metrics.json`.

4.2. RQ1–RQ2: Latent forecast and data scale

Table 3 and Figure 3 are the headline forecast result. Causal L1 exceeds persistence, query-token L1, Mamba, and GRU on the same 750-clip val set, with the lowest MSE. The margin over GRU is small in absolute cosine (0.004) but decisive in a paired test: at $h=1$, 0.9811 vs 0.9777 (Holm $p=1.24 \times 10^{-84}$); at $h=4$, 0.9776 vs 0.9740 ($p=4.06 \times 10^{-102}$). Against Mamba the $h=4$ Holm p is 1.23×10^{-111} . Predictor *form* is the lever: query-token L1 (70M) loses to GRU (2.6M); switching the same Transformer to causal autoregression plus residual (4) reverses the ranking.

The scale curve (Figure 3a) is monotonic from 500 to 6,000 clips ($0.962 \rightarrow 0.968 \rightarrow 0.972 \rightarrow 0.976 \rightarrow 0.978$) and stays above the 6k GRU throughout. Extending the same causal L1 to all 13,552 pooled clips that the local corpus yields leaves cosine flat at 0.978 and trims MSE from 0.139 to 0.135 (val $n=1,631$). The model is data-scalable up to a few thousand clips; beyond that, on this frozen encoder, forecast cosine has reached a plateau and further gains should come from method (dense spacetime L1, action supervision), not from stacking more of the same pooled tokens.

Pixel-generation baselines confirm Proposition 1: a next-frame CNN reaches 17.9 dB PSNR against copy-last 29.6 dB; a conditional DDPM reaches 4.9 dB against copy-last 30.4 dB (Figure 6a). Pixel error concentrates on unpredictable appearance; representation-space forecast does not have to.

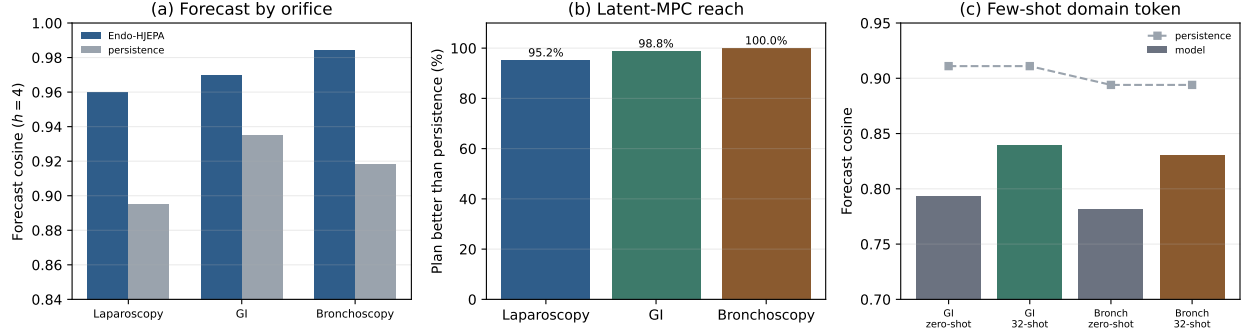


Figure 4: (a) Forecast by orifice on the 2,000-clip full model (same protocol as Table A.7). (b) Latent-MPC reach versus persistence. (c) Few-shot domain-token adaptation (32 target clips) from a laparoscopy-only model. Source: p2000_full_causal/eval_ckpt.json, t16_transfer_laparo/fewshot_*.json.

4.3. RQ3: Latent planning and physical energy

Planning uses L3+energy+MPC and has no GRU counterpart. On the 2,000-clip full model (250 val clips), latent MPC is closer to the goal latent than persistence on **98.0%** of clips (cosine 0.920 vs 0.874). By orifice: laparoscopy 95.2%, GI 98.8%, bronchoscopy 100% (Figure 4b). Planned actions have lower energy than random actions on 90.8% of clips. On SCARED depth, energy flags near-wall transitions above chance (AUC 0.682, Spearman -0.471 , $n=132$; Figure 6b). This is the physical reading of the energy prior: it correlates with camera-to-tissue distance, the quantity that matters for wall-contact safety.

Goal-directed navigation to an *arbitrary* SCARED viewpoint is a harder task: latent reach is 10% of 10 trials, with no pose-error reduction versus persistence. The world model predicts the *likely* future well; it cannot yet steer to an arbitrary anatomical target, because latent actions do not encode how to move (Section 4.5). We report this as a boundary, not as a success.

4.4. RQ4: Cross-orifice transfer

A laparoscopy-only model beats persistence in-domain (0.922 vs 0.850) but falls *below* persistence on zero-shot GI (0.791 vs 0.913) and bronchoscopy (0.776 vs 0.893). Fine-tuning only the domain embedding on 32 target clips recovers +0.045 (GI) and +0.049 (bronch) (Figure 4c). The unified model beats persistence in every domain (Table A.7). Orifice domains are distinct enough that zero-shot transfer fails; unified training, or a cheap domain token, is necessary.

4.5. RQ5: Action grounding (negative result)

Emergent latent actions are not physically grounded (SCARED pose NMI 0.41–0.52 versus random ≈ 0.40 ; residual $\rightarrow$ twist R^2 negative) and not semantically grounded (CholecT50 verb NMI 0.044 versus random 0.056; probe accuracy 0.306 versus majority-class 0.390). A codebook-level verb loss does not help. Encoder-level verb supervision lifts residual $\rightarrow$ verb NMI from chance 0.026 to 0.056—above chance, still weak. Latent actions index visual change that is useful for forecast and MPC; they are not discovered surgical actions.

Table 4: CholecT50 official split, video-level linear probe, mean $\pm$ std over 3 seeds. Source: `outputs/vjepa2_adapted/cholect50_probe_multiseed.json`. We do not claim uniform dominance.

Encoder	Phase acc	Instrument mAP
ImageNet ViT (supervised)	0.592 ± 0.016	0.430 ± 0.003
DINOv2 (image SSL)	0.658 ± 0.016	0.575 ± 0.003
VideoMAE (Kinetics SSL)	0.558 ± 0.012	0.497 ± 0.004
TimeSformer (video ViT)	0.683 ± 0.012	0.490 ± 0.010
ViViT (video ViT)	0.692 ± 0.024	0.443 ± 0.004
V-JEPA 2 (frozen)	0.704 ± 0.006	0.406 ± 0.002
V-JEPA 2 (domain-adapted)	0.688 ± 0.057	0.485 ± 0.005

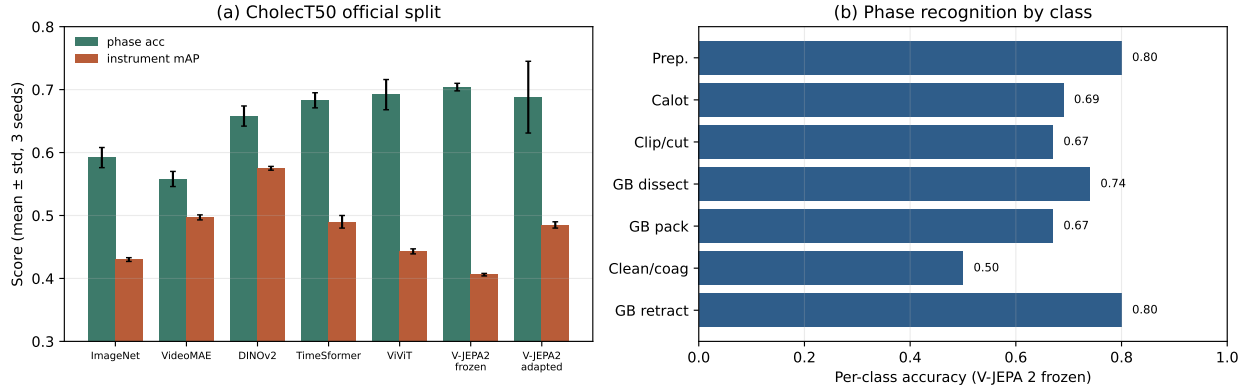


Figure 5: (a) Official-split probes matching Table 4 (error bars = 3-seed std). (b) Per-class phase accuracy of frozen V-JEPA 2: cleaning/coagulation is hardest (0.50), preparation and retraction easiest (0.80).

4.6. RQ6: Downstream recognition

On the official challenge videos (VID68, 70, 73–75), frozen V-JEPA 2 is the strongest *video* model for phase (0.704), above ViViT, TimeSformer, DINOv2, ImageNet and VideoMAE. DINOv2 leads instrument presence (0.575); our adaptation improves instrument mAP over frozen V-JEPA 2 ($0.406 \rightarrow 0.485$) at a small, high-variance phase cost. This is an encoder-quality table, not a claim that the world model is a recognition SOTA. STIR fine-tuning reduces held-out deformation chamfer from 179.9 to 168.2 (−6.5%; Figure 6c).

4.7. Per-dataset decomposition

“Per-domain” (three orifices) hides dataset-level variation. We reconstruct the same 750-clip video-level val list used to build the 6,000-clip cache (`domain_balanced_indices`, seed 1), attach dataset names, and evaluate the causal-L1 checkpoint. Alignment is exact ($n_{\text{cache}}=n_{\text{rebuild}}=750$). Table 5 and Figure 8 report every dataset with $n \geq 5$; SCARED and STIR appear with $n=1$ in the JSON and are omitted from the table.

The unified model beats persistence on every listed dataset. The largest margins are Stereo_Lap (+0.093) and ION bronchoscopy (+0.071), consistent with those videos being camera-motion-dominated and therefore the most predictable in latent space. Kvasir-Capsule has the strongest persistence (0.941) and the smallest—but still positive—margin (+0.037): capsule motion is slower, so copying the last token is already a good forecast. This is the

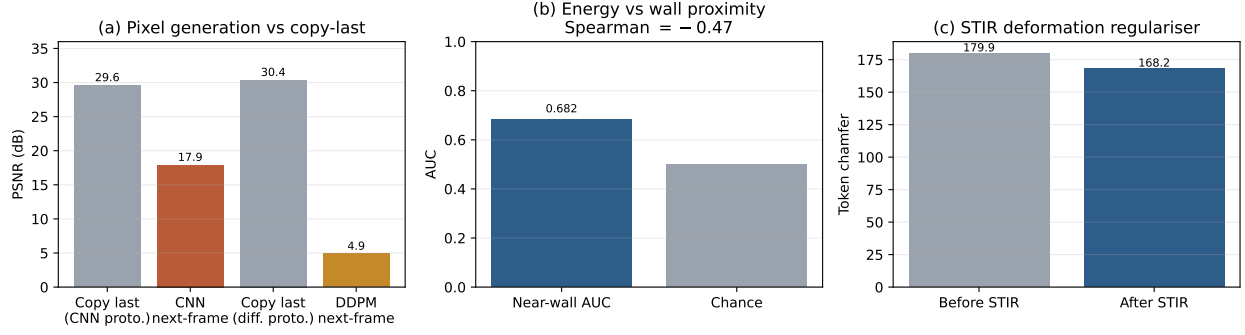


Figure 6: (a) Pixel world-model baselines lose to copy-last, consistent with Proposition 1. (b) Energy versus SCARED wall proximity. (c) STIR token chamfer before/after the regulariser (8).

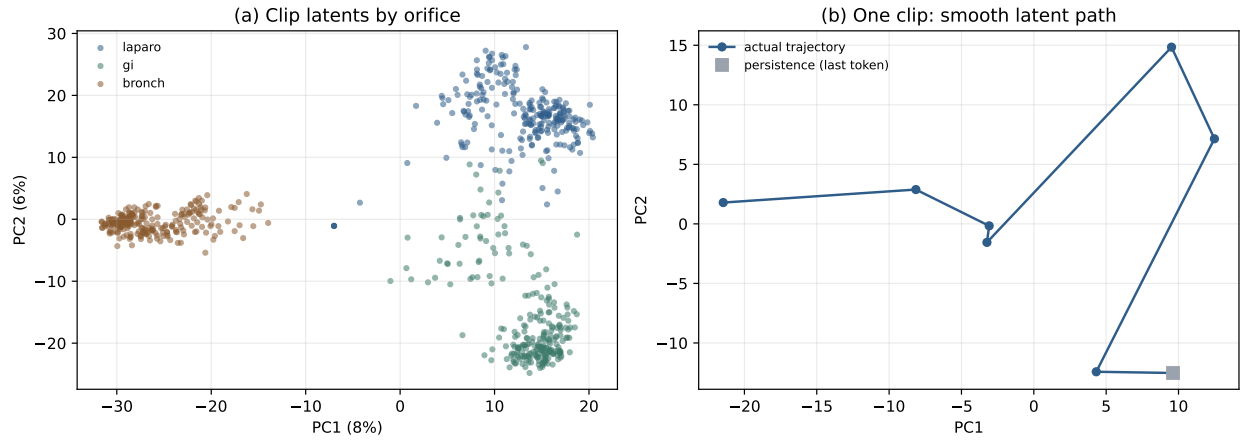


Figure 7: (a) PCA of pooled val latents coloured by orifice: a partially shared space with unsupervised domain structure. (b) One clip’s latent trajectory is smooth and low-dimensional, which is why persistence is a strong short-horizon baseline.

dataset-level reading of the paper’s claim: unification helps, and it does not secretly rely on one easy corpus.

4.8. Ablations

Three findings. (i) Causal autoregression is the single largest forecaster improvement ($0.936 \rightarrow 0.978$). (ii) Residual anchoring is essential at modest scale: without it the large predictor drops below persistence on its own cache (0.768 vs 0.886). (iii) L2/L3 add little to short-horizon *forecast* (full H-JEPA 0.971 at $2k$) but are what enable planning and the energy signal. Hyperparameter sensitivity on the $6k$ val is <0.002 .

5. Discussion

5.1. Clinical significance

The target is anticipatory endoscopic intelligence, not another phase-recognition point. Forecast is the computational primitive behind collision warning and loss-of-view recovery. The energy head is the first quantity in this stack that correlates with a physical safety signal

Table 5: Per-dataset latent forecast (6,000-clip causal L1, $h=4$, video-level val). Source: `outputs/scale_6000_causal/per_dataset.json (aligned=true)`. Overall 0.978 / 0.916 matches Table 3.

Dataset	Domain	n	cos	persist	Δ
ION_bronch	bronch	250	0.988	0.918	+0.071
Kvasir-Capsule	gi	206	0.978	0.941	+0.037
CholecT50	laparo	89	0.963	0.894	+0.069
Stereo_Lap	laparo	75	0.982	0.889	+0.093
endoscapes	laparo	72	0.961	0.893	+0.068
HyperKvasir	gi	44	0.970	0.924	+0.046
endovis2018_full	laparo	7	0.961	0.882	+0.079
endovis2017_full	laparo	5	0.965	0.909	+0.056
All val clips	—	750	0.978	0.916	+0.062

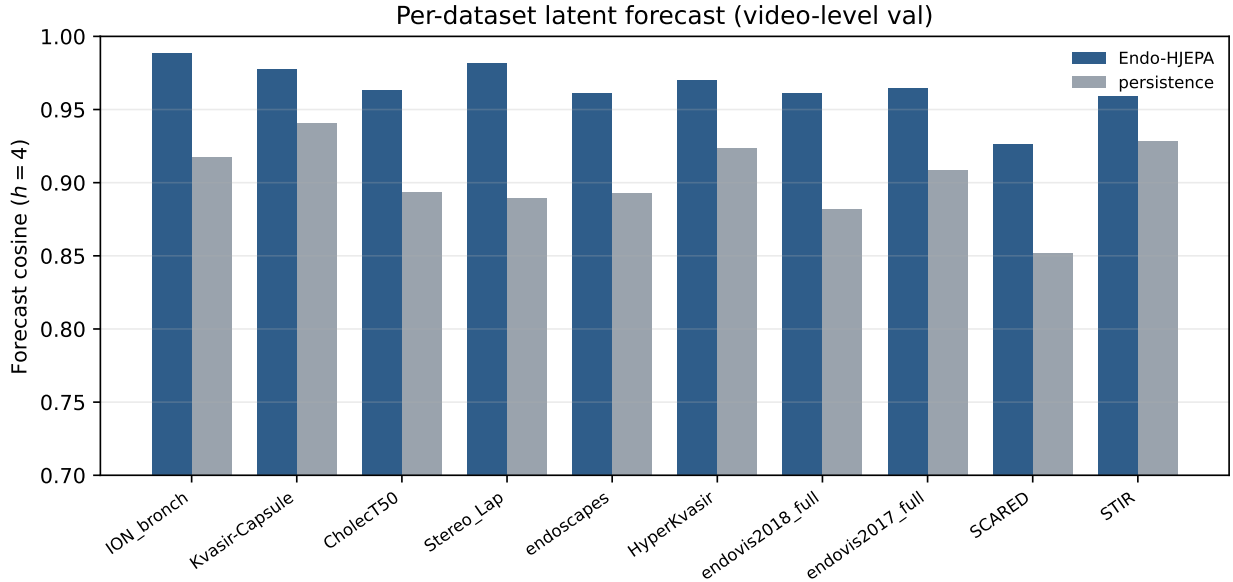


Figure 8: Per-dataset forecast corresponding to Table 5 (plus $n=1$ rows SCARED/STIR, not used in the table). The model beats persistence on every dataset; the largest margins are camera-motion-dominated corpora (Stereo_Lap, ION).

(camera-to-tissue distance). Unification is the data-efficient path for bronchoscopy, where labelled video is scarce. All of this remains in-silico.

5.2. What the method actually contributes

The innovations that move the numbers are, in order: residual anchoring (keeps a large predictor above persistence), causal L1 (beats GRU/Mamba), domain tokens (recover failed zero-shot transfer), and the energy head (enables MPC and tracks wall proximity). Hierarchy (L2/L3) is not a short-horizon cosine trick; it is the interface that makes planning well-posed. That is why we refuse to sell full H-JEPA as a 0.003 forecast win over L1-only.

Table 6: Design ablations. The first five rows share the 6,000-clip protocol of Table 3. Residual on/off uses a different val cache (persist 0.886) and is not mixed into the 0.916 persist row.

Variant	cos $h=4$	Protocol
Persistence	0.916	6k shared cache
Query-token L1	0.936	6k
Mamba / SSM	0.971	6k
GRU	0.974	6k
Causal L1	0.978	6k
Full H-JEPA (L1+L2+L3+E)	0.971	2k full, forecast only
w/o residual (other cache)	0.768	persist 0.886 on that cache
with residual (same cache)	0.900	persist 0.886

5.3. Failure analysis

A small GRU is nearly competitive at $h=1$. Goal-directed navigation to arbitrary targets fails because latent actions encode visual change, not steer commands. Zero-shot GI/bronch transfer fails because the domains are genuinely distinct at the dynamics level. Each failure points to a concrete remedy (action supervision, domain-token adaptation) rather than to an overclaim.

5.4. Comparison to published world models

Endo-HJEPA occupies a distinct design point: hierarchical + energy-based + planning, unified across orifices, no pixel decoder. V-JEPA 2-AC, SurgRec-JEPA, EndoWAM and SVWM do not release public endoscopic-planning checkpoints, so we compare against the baselines we can run (Table 3, Table 4) and cite the others as related work, not as a fabricated leaderboard.

5.5. Limitations

(i) Training scale is far below SurgRec ($\sim 2 \times 10^8$ frames). (ii) Full Cholec80 and extra C3VD trajectories depend on access / Drive quota. (iii) ION has no public telemetry. (iv) Latent actions are not interpretable. (v) All planning is in-silico. (vi) Val forecast is domain-balanced, so per-dataset n is not proportional to corpus size; $n=1$ rows are omitted.

6. Conclusion

Endo-HJEPA brings hierarchical, energy-based joint-embedding world modelling to unified cross-orifice endoscopic video. A causal residual forecaster surpasses persistence, GRU and Mamba on a leakage-free protocol; the same stack is the only compared model that performs energy-guided latent planning; the energy head correlates with physical wall proximity; and a per-dataset decomposition shows the forecast win is not carried by a single corpus. Latent actions remain a planning device. We release the protocol, the verified metric file, and the figure generators so that every number in this paper can be regenerated from outputs/.

Declarations

Ethics. Public datasets are used under their original licences; ION cases are anonymised (case_XXX); an ethics approval number will be added before publication. All planning evaluation is in-silico. **Competing interests.** None declared. **Data/code availability.** Public datasets from their sources; the `endoworld` package, `verified_metrics.json`, and figure scripts are released with this manuscript.

Appendix A. Per-domain planning (2,000-clip full H-JEPA)

Table A.7: Forecast and planning by orifice (full H-JEPA, 2,000-clip, 250 val clips). Source: `p2000_full_causal/eval_ckpt.json`. Do not mix with Table 3 (6k causal L1).

Domain	Forecast cos (model / persist)	Plan reach (%)	n
Laparoscopy	0.960 / 0.895	95.2	84
GI endoscopy	0.970 / 0.935	98.8	83
Bronchoscopy	0.984 / 0.918	100.0	83

Appendix B. Per-class phase recognition

Table B.8: Per-class phase accuracy (V-JEPA 2 frozen, official split, 3-seed mean).

Phase	Acc
Preparation	0.80
Calot triangle dissection	0.69
Clipping & cutting	0.67
Gallbladder dissection	0.74
Gallbladder packaging	0.67
Cleaning & coagulation	0.50
Gallbladder retraction	0.80

Appendix C. Reproducibility

Forecast: `endoworld.world.train` and `endoworld.eval.per_dataset`. Planning / energy: `endoworld.eval.eval_ckpt`, `scared_collision`, `scared_navigation`. Recognition: `endoworld.eval.cholect50_probe`. Figures: `docs/endohjepa/make_figures.py` reads only `verified_metrics.json`. Encoder caches are shared across ablations; probe heads use a fixed seed.

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